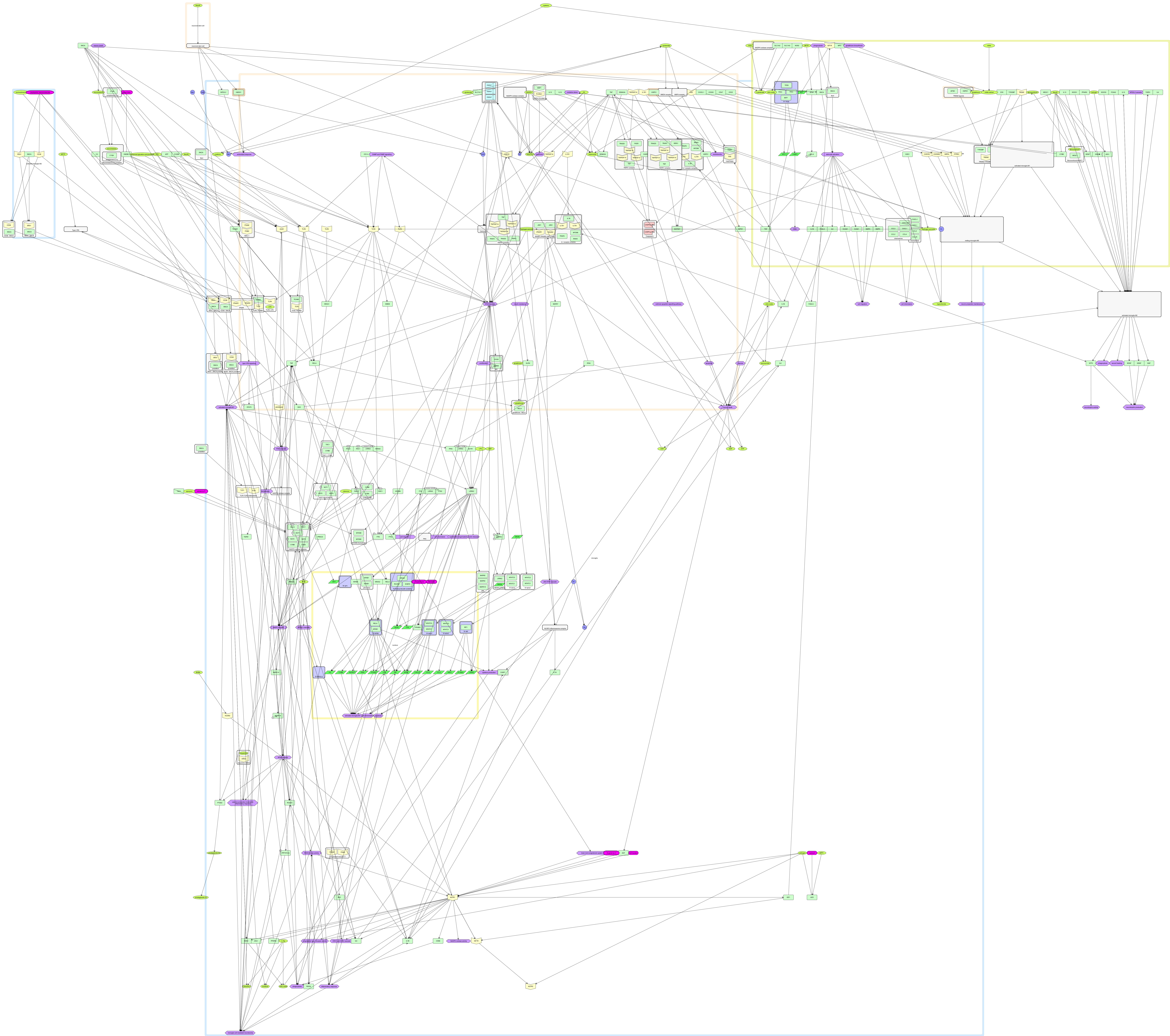
mitochondrion
HSPD1
neuron
CD22
CD200
CX3CL1
CD47
SEMA3A
NRP1
PLXNA1
NXPH1:PLXNA1
apoptosis
PTPRC
CD200R1
CX3CR1
SIRPA
neuron projection maintenance
resting microglia M0
IL13
IL10
activated microglia M1
TREM2
SYK
TYROBP
TREM2:TYROBP:SYK
APOE
TREM2 ligands
apoptotic neuron
glucocorticoid
NR3C1
Glucocorticoid:NR3C1
NFE2L2 cascade
IL4
TGFB1
FOXA2
ARG1
SOCS3
SOCS1
PPARG
naringin
fasudil
GDNF
NOS2
CYBB
GSK3B
oxidative stress
neurotrophin production
BDNF
NTF3
NGF
activated microglia M2
wound healing
phagocytosis
SHH
NR4A2
PI3K cascade
Akt pathway
microglia cell mediated neurotoxicity
purmorphamine
SRA1
SNCA
SRA1 : SNCA
protofibril
SRA1 : SNCA protofibril
CD36
CD36 : SNCA protofibril
CD36 : SNCA
amphiphilic macromolecules
HMGB1
fibril
APP
LPS
TLR3
TLR2
CD14
TLR4
DAMP and PAMP signalling
neuron death
ATP
ADP
Na+
Ca2+
K+
UDP
P2RX7
purinergic receptor
P2RY6
IL1R1
IL1B
CASP1
Calcineurin
NLRP3 inflammasome complex
hydrogen peroxide
ROS
NFkB cascade
NFE2L2 cascade
MAPK cascade
CHUK
IKBKB
IKK
LRRK2
inflammatory response
cytoskeleton organisation
cell migration
cell chemotaxis
IFNG
tat HIV
mutated G2019S
PTK2
mutated R1441G
PKA
NFKB1
p50 dimer
RELA
TF NFKB
TF NFE2L2
TF AP1
IRF1
TF IRF
NRP1
PLXNA1
MAPK7
JAK-STAT cascade
JAK1
JAK2
IFNGR1
IFNGR2
IFNGR1:IFNGR2:JAK1:JAK2
S100B
AGER
advanced glycation end-products
neuromelanin
C1QA
complement C1q
Neuromelanin C1q complex
ITGAM
ITGB2
MAC-1
C3
ADORA2A
rosiglitazone
MDR548
nucleus
NLRP3
IL6
PTGS2
TNF
IL18
cytokine production
TRAF6
IRAK4
MYD88
IL1 receptor complex
TRADD
TRAF2
RIPK1
TNFRSF1A
TNFR1 complex
MAP3K7
MPTP
Type I IFN
IFNAR1
IFNAR2
IFNAR
TLR4: LPS
FADD
extrinsic apoptotic signalling pathway
CASP8
CASP8 (p10)
Caspase 8
FAS
FAS:FADD
FASLG
hypoxia
peroxynitrite
nitric oxide
prostaglandin E2
L-Arg
citrulline
ornithine
SOD1
superoxide
O2
NADPH oxidase complex
NADPH
NADP(+)
H+
TAC1
TAC1 : CYBB
NCF1
NCF2
NCF4
NCF1:NCF2:NCF4
CYBA
CYBA:CYBB
RAC1
RAC2
GDP
GTP
proliferation
type I IFN signaling
mutated T2379M
STAT3
Crohn disease
NFATC3
NFATC1
NFATC2
TF NFAT
NRON
NRON complex
IRF7
CCL3
CCL2
CXCL2
cell chemotaxis
NR4A2
RCOR1
EHMT2
TF NR4A2:RCOR1:EHMT2
IKBKB
FYN
GLRX
insulin resistance
glutathione
glutathione : RELA
MYD88 Homodimer
MAPK8
MAPK9
MAPK10
JNK
PRKCD
MAPK14
NR4A1
TICAM1
TLR2: TICAM1
TLR4: TICAM1
TLR1
TLR1:TLR2 heterodimer
TLR9
microglia
substance P
apocynin
ITGAM
ITGB2
complement receptor 3
rotenone
SRC
ERK1 and ERK2 cascade
RHOA
MAPK14
ROCK2
excitotoxicity
NMDA receptor
AMPA receptor
L-glutamate
SLC7A11
cysteine
SLC1A2
SLC1A3
glutathione biosynthesis
glutathione
hypochloride
nitrite radical
Cl-
nitrite
MPO
Angiotensin II
AGT
TFEB
TFEC
TFE3
MITF
TF TFEB
NPC1
LAMP2
CMA
brain renin-angiotensin system
IGF1R
IGF1
AGTR1
AGTR2
dopamine
DRD2
dopamine: DRD2
glycyrrhizin
CD68
NADPH oxidase activity
cytokine production involved in inflammatory response
HCAR2
BHBA
PPP1R12A
antioxidant response
NFE2L2
HMOX1
ER
neurotrophic activity
fasudil
neuronal stem cell
GFAP
MPTP
chemokines
CXCL2
CXCL1
CCL4
CX3CL1
CCL20
astrocyte activation
angiotensin type I receptor activity
estrogen
ACE
MPP+
captopril
ROCK kinase activity
telmisartan
ICAM1
VCAM1
MMP9
MMP3
cell migration
astrocyte


fasudil
neuronal stem cell
neuronal stem cell
cysteine
SNCA
neuron death
purmorphamine
amphiphilic macromolecules
neuromelanin
C1QA
complement C1q
glycyrrhizin
Na+
Ca2+
NFE2L2
HMOX1
L-glutamate
SLC7A11
P2RX7
P2RX7
P2RX7
purinergic receptor
NADPH oxidase complex
NADPH
NRP1
PLXNA1
NRP1:PLXNA1
IL13
IL10
oxidative stress
O2
TNF
SEMA3A
TNFRSF1A
IL1R1
HSPD1
NMDA receptor
AMPA receptor
FAS
CX3CL1
CD200
CD47
CD22
ROS
NADPH oxidase complex
SLC1A2
SLC1A3
NOS2
MPTP
phagocytosis
IGF1R
MPO
glutathione biosynthesis
nitrite
L-glutamate
nitric oxide
TFEB
TFEC
TFE3
MITF
TF TFEB
AGT
GFAP
SNCA
SNCA
fibril
APOE
HSPD1
TREM2 ligands
glutathione
nitrite radical
SYK
TYROBP
TREM2
glucocorticoid
NR3C1
fasudil
IL13
SOCS1
PPARG
naringin
SOCS3
FOXA2
IL10
NFE2L2 cascade
TGFB1
IL4
SRA1
SNCA
CD36
activated microglia M2
MPTP
C3
neuromelanin
C1QA
complement C1q
Neuromelanin C1q complex
HMGB1
advanced glycation end-products
LPS
APP
S100B
fasudil
SNCA
fibril
cysteine
Na+
antioxidant response
CD14
DAMP and PAMP signalling
Ca2+
P2RY6
H+
NADP(+)
apoptosis
TNFRSF1A
IL1R1
superoxide
SEMA3A
TRADD
FADD
TNFRSF1A
TNFRSF1A
TNFRSF1A
TNF
TNFR1 complex
TRADD
TRAF2
RIPK1
TNFRSF1A
TNFRSF1A
TNFRSF1A
TNF
TNFR1 complex
IRAK4
MYD88
TRAF6
IL1R1
IL1R1
IL1B
IL1 receptor complex
HSPD1
excitotoxicity
FADD
FAS
FAS:FADD
NPC1
LAMP2
RELA
astrocyte activation
SOD1
CX3CR1
CD200R1
SIRPA
PTPRC
TYROBP
TREM2
TREM2:TYROBP
activated microglia M1
CYBB
glucocorticoid
NR3C1
Glucocorticoid:NR3C1
GDNF
GSK3B
ARG1
CD36
SNCA
CD36 : SNCA
SRA1
SNCA
SRA1 : SNCA
Type I IFN
TICAM1
ITGAM
ITGB2
MAC-1
AGER
TLR2
TLR9
TLR4
TLR4
TLR3
Calcineurin
TNF
TNFRSF1A
TNFRSF1A
TNFRSF1A
TRAF2
TRADD
RIPK1
TNFR1 complex
hydrogen peroxide
JAK1
JAK2
IFNGR1
IFNGR2
IFNGR1:IFNGR2:JAK1:JAK2
IL1B
IL1R1
IL1R1
MYD88
TRAF6
IRAK4
IL1 receptor complex
CASP8 (p18)
CASP8 (p10)
Caspase 8
MAP3K7
HSPD1
TNF
CMA
IL1B
FASLG
IL6
VCAM1
ICAM1
MMP3
MMP9
CCL3
CXCL10
CXCL1
CCL2
CCL4
chemokines
CX3CL1
CXCL1
CCL2
CCL20
chemokines
hydrogen peroxide
Cl-
resting microglia M0
SRA1
SNCA
SRA1 : SNCA
CD36
SNCA
CD36 : SNCA
IFNAR1
IFNAR2
IFNAR
TICAM1
TLR4
TLR4: TICAM1
TLR4
LPS
TLR4: LPS
TICAM1
TLR2
TLR2: TICAM1
NR4A1
IKBKB
IKBKB
NFkB cascade
insulin resistance
MAPK7
extrinsic apoptotic signalling pathway
nitric oxide
IL1B
nitric oxide
IL1B
FASLG
cell migration
cell chemotaxis
hypochlorite
neuron projection maintenance
SRA1
SNCA
protofibril
SRA1 : SNCA protofibril
CD36
SNCA
protofibril
CD36 : SNCA protofibril
type I IFN signalling
TNF
RELA
proliferation
CHUK
IKBKB
IKK
glutathione
GLRX
IFNG
apoptosis
hypoxia
peroxynitrite
IL6
peroxynitrite
IL6
NTF3
phagocytosis
wound healing
BDNF
GDNF
NGF
neurotrophic activity
neurotrophin production
activated microglia M1
STAT3
ADORA2A
SHH
glutathione
RELA
glutathione : RELA
neuron death
UDP
ADP
ATP
SNCA
protofibril
PI3K cascade
TAC1
CYBB
TAC1 : CYBB
RAC2
RAC1
mutated R1441G
LRRK2
NR4A2
IFNG
mutated T2356I
LRRK2
tat HIV
LPS
ADP
TAC1
apocynin
substance P
TLR1
TLR2
TLR1:TLR2 heterodimer
Akt pathway
NADPH oxidase complex
NCF1
NCF2
NCF4
NCF1:NCF2:NCF4
rotenone
RAC2
CYBB
CYBA
CYBA:CYBB
RAC1
MYD88
FYN
mutated G2019S
LRRK2
PTK2
Y397
LRRK2
RAC2
RAC1
NCF1
NCF2
NCF4
CYBB
CYBA
NADPH oxidase complex
TGFB1
PRKCD
MYD88
MYD88
MYD88 Homodimer
FYN
PTK2
cell migration
PKA
cell chemotaxis
cytoskeleton organisation
Crohn disease
S935
S910
LRRK2
NRON
microglia
PRKCD
ROS
IRF7
TF AP1
NFKB1
NFKB1
NFKB1
p50 dimer
NR4A2
RELA
NR4A2
RCOR1
EHMT2
TF:NR4A2:RCOR1:EHMT2
fingolimod
MDG548
MAPK8
MAPK9
MAPK10
JNK
LRRK2
NRON
NRON complex
NFATC3
NFATC1
TF NFAT
NFATC2
NFATC1
TF NFAT
JAK-STAT cascade
K+
MAPK cascade
NFE2L2 cascade
RELA
NFKB1
TF NFKB
GLRX
FYN
PPARG
NFATC3
NFATC1
TF NFAT
NFATC2
NFATC1
TF NFAT
IRF1
TF IRF
NLRP3 inflammasome complex
K+
nucleus
MAPK14
TF NFE2L2
IL6
CYBB
NLRP3
IFNG
PTGS2
IL1B
TNF
NOS2
CXCL2
CCL3
CCL2
NRP1
PLXNA1
LRRK2
cytokine production
CASP1
IL18
activated microglia M1
cell chemotaxis
apoptosis
BHBA
HCAR2
MAPK14
dopamine
DRD2
dopamine: DRD2
NFkB cascade
PTGS2
cytokine production involved in inflammatory response
ROCK2
prostaglandin E2
PPP1R12A
ROCK kinase activity
ITGAM
ITGB2
complement receptor 3
prostaglandin E2
SRC
brain renin-angiotensin system
Angiotensin II
AGT
lisinopril
estrogen
captopril
MPP+
AGTR1
IGF1
ACE
IGF1
NOS2
ARG1
PPARG
L-Arg
angiotensin type I receptor activity
ERK1 and ERK2 cascade
IL6
IL6
IL1B
CD68
NADPH oxidase activity
IGF1R
AGTR2
citrulline
ornithine
nitric oxide
phagocytosis
RHOA
inflammatory response
microglia cell mediated neurotoxicity